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Meta-Analysis

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The use of statistical methods to combine the results of independent empirical research studies (meta-analysis) has a long history. Meta-analytic work can be divided into two traditions: tests of the statistical significance of combined results and methods for combining estimates across studies. The principal classes of combined significance tests are reviewed, and the limitations of these tests are discussed. Fixed effects approaches treat the effect magnitude parameters to be estimated as a consequence of a model involving fixed but unknown constants. Random effects approaches treat effect magnitude parameters as if they were sampled from a universe of effects and attempt to estimate the mean and variance of the hyperpopulation of effects. Mixed models incorporate both fixed and random effects. Finally, areas of current research are summarized, including methods for handling missing data, models for publication selection, models to handle studies that are not independent, and distribution-free models for random effects.

NEW APPLICATIONS OF META-ANALYSIS TO SOME OLD STATISTICAL PROBLEMS

The term *meta-analysis* was introduced by Gene Glass in his 1976 presidential address to the American Educational Research Association as "the analysis of the results of statistical analyses for the purposes of drawing general conclusions" (p. 3). Since the introduction of the term, the use of statistical methods to combine the results of replicated research studies has become widespread in education, psychology, and the biomedical sciences. Although the term was new in 1976, the actual use of statistical methods to combine the results of a series of experiments has a long history.

Early Examples of Meta-Analysis

One of the earliest examples of combining information from several research studies for the purposes of drawing a general conclusion arises in 19th century astronomy and geodesy (see Stigler, 1986). In these studies, the result of each study was a series of astronomical observations. The studies were typically conducted at different sites (observatories) and under different conditions. The parameters of scientific interest were the unknowns of a system of linear equations in which the observations were the (known)

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coefficients. (Today, we would characterize these unknowns as the coefficients of a linear regression equation.) Stigler credits Legendre with the invention of the principle of least squares to solve the problem of how to estimate these coefficients. One could equally well point to Legendre as an innovator in meta-analysis because his purpose was to combine information across studies. Methods similar to Legendre's continue to be used to combine estimates of physical constants (Birge, 1932).

Another early example of meta-analysis was a study by Karl Pearson (1904) of the effectiveness of inoculation for enteric (typhoid) fever. Pearson presented five sets of data consisting of 2×2 tables giving the rates of infection for samples of people who were inoculated and for those who were not. Each set of data was collected separately under different conditions and in a different geographical location (India, South Africa, etc.). Pearson computed a tetrachoric correlation from each table as an index of the relationship between inoculation and infection. After examining the variation among the five correlations, Pearson computed the average of the correlations to summarize the "typical" value of the relationship. He used the average correlation as an index of the size of the effect of the treatment (inoculation). In fact, he compared it to the typical size of the effect for smallpox inoculation and concluded that the enteric fever inoculation was not as effective.

Pearson's early meta-analysis has all the features of a contemporary meta-analysis. He computed an index of effect that would be comparable across studies. He examined the degree of homogeneity of effects across studies, taking into account the sampling uncertainty of the estimate from each study (in the form of the probable error of each correlation). Once he assured himself of the homogeneity of the correlations across studies, he computed an average effect (average correlation) and used the average as a representative of the effect of the treatment.

Methodological Work

The first written work dealing specifically with the methodology of combining the results of experiments does not appear until the 1930s. At this point, two distinct traditions of work emerge. One tradition concerns the combination of hypothesis tests, and the other concerns combination of estimates of treatment effects.

Combining Hypothesis Tests

One of the first proposals for a test of the statistical significance of combined results (now called testing the minimum p or Tippett method) was given by L. H. C. Tippett in 1931. Soon afterwards, R. A. Fisher (1932) proposed a method for combining tests of statistical significance based on the product of the p values across studies. Karl Pearson (1933) indepen-

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dently derived the same method shortly thereafter. Research on tests of the significance of combined results has flourished since that time, and now well over 100 papers in the statistical literature have been devoted to such tests.

The problem that motivates combining hypothesis tests across studies was stated best by R. A. Fisher:

> When a number of quite independent tests of significance have been made, it sometimes happens that although few or more can be claimed individually as significant, yet the aggregate gives an impression that the probabilities are on the whole lower than would often have been obtained by chance. It is sometimes desired, taking account only of these probabilities, and not of the detailed composition of the data from which they are derived, which may be of very different kinds, to obtain a single test of the significance of the aggregate. (1932, p. 99)

Tests of the significance of combined results are sometimes called omnibus or nonparametric tests because these tests do not depend on the type of data or the statistical distribution of those data. Instead, tests of the statistical significance of combined results rely only on the fact that, under the null hypothesis, p values are uniformly distributed between zero and unity. Although omnibus tests have strong appeal in that they can be applied universally, they suffer from an inability to provide estimates of the magnitude of the effects being considered. Thus, omnibus tests do not provide information about the size of the treatment effect.

Consider a collection of k independent studies characterized by parameters $\theta_1, \ldots, \theta_k$, such as means, mean differences, or correlations. Assume further that the ith study produces a continuous test statistic T_i used to test the null hypothesis

$$H_{0i}\colon\ \theta_i = 0, \qquad i = 1, \ldots, k,$$

where large values of the test statistic lead to rejection of the null hypothesis. The hypotheses $H_{01}, \ldots, H_{0k}$ need not have the same substantive meaning, and, similarly, the statistics $T_1, \ldots, T_k$ need not be of related form. The omnibus null hypothesis H_0 is that all the θs are zero:

$$H_0\colon\ \theta_1 = \theta_2 = \cdots = \theta_k = 0.$$

Note that the composite hypothesis H_0 holds only if each of the subhypotheses $H_{01}, \ldots, H_{0k}$ holds.

The one-tailed p value for the ith study is

$$p_i = \text{Prob}\{T_i > t_{i0} \mid \theta_i = 0\}, \tag{1}$$

where t_{i0} is the value of the statistic actually obtained (the sample realization of T_i) in the ith study. If H_{0i} is true, the p_i is uniformly distributed on the interval $[0, 1]$.

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The Minimum p *Method*

The first test of the significance of combined results was proposed by Tippett (1931), who pointed out that, if $p_1, \ldots, p_k$ are independent p values (from continuous test statistics), then each has a uniform distribution under H_0. Therefore, if $p_{[1]}$ is the minimum of $p_1, \ldots, p_k$, a test of H_0 at significance level α is obtained by comparing $p_{[1]}$ with $1 - (1 - \alpha)^{1/k}$, so that the test procedure is to

$$\text{reject } H_0 \text{ if } p_{[1]} < 1 - (1 - \alpha)^{1/k}. \tag{2}$$

The Inverse Chi-Square Method

Perhaps the most widely used combination procedure is that of Fisher (1932). Given k independent studies and the p values $p_1, \ldots, p_k$, Fisher's procedure uses the product $p_1 p_2 \cdots p_k$ to combine the p values. He made use of a connection between the uniform distribution and the chi-square distribution to show that, under H_0, $-2 \log p$ has a chi-square distribution with 2 degrees of freedom. Because the sum of independent chi-square variables has a chi-square distribution, we have the very simple and elegant fact that, if H_0 is true, then

$$-2 \log(p_1 p_2 \cdots p_k) = -2 \log p_1 - \cdots - 2 \log p_k$$

has a chi-square distribution with $2k$ degrees of freedom. Because of this fact, no special tables are needed for the Fisher method. The test procedure becomes

$$\text{reject } H_0 \text{ if } P = -2 \sum_{i=1}^{k} \log p_i > C, \tag{3}$$

where the critical value C is obtained from the upper tail of the chi-square distribution with $2k$ degrees of freedom.

The Inverse Normal Method

Another procedure for combining p values is the inverse normal method proposed by Stouffer, Suchman, DeVinney, Star, and Williams (1949). This procedure involves transforming each p value to the corresponding normal score and then "averaging." More specifically, define z by $p_i = \Phi(z_i)$, where $\Phi(x)$ is the standard normal cumulative distribution function. Where H_0 is true, the statistic

$$Z = \frac{z_1 + \cdots + z_k}{\sqrt{k}} = \frac{\Phi^{-1}(p_1) + \cdots + \Phi^{-1}(p_k)}{\sqrt{k}} \tag{4}$$

has the standard normal distribution. Hence we reject H_0 whenever Z exceeds the appropriate critical value of the standard normal distribution.

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The Logit Method

Yet another method for combining k independent p values $p_1, \ldots, p_k$ was suggested by George (1977) and investigated by Mudholkar and George (1979). It involves transforming each p value into a logit, $\log[p/(1-p)]$ and then combining the logits via the statistic

$$L = \log\frac{p_1}{1-p_1} + \cdots + \log\frac{p_k}{1-p_k}.$$

The exact distribution of L is not simple, but, when H_0 is true, George and Mudholkar (1977) show that the distribution of L (except for a constant) can be closely approximated by Student's t distribution with $5k + 4$ degrees of freedom. Therefore, the test procedure using the logit statistic is

$$\text{reject } H_0 \text{ if } L^* = |L|\sqrt{3(5k+4)/k\pi^2(5k+2)} > C, \tag{5}$$

where the critical value C is obtained from the t distribution with $5k + 4$ degrees of freedom.

Limitations of Combined Significance Tests

In spite of the intuitive appeal of using nonparametric procedures to combine tests of treatment effects, there frequently are problems in the interpretation of results of such a test of the significance of combined results (e.g., see Adcock, 1960; Wallis, 1942). Just what can be concluded from the results of the significance of combined results? Recall that the null hypothesis of the combined test procedure is

$$H_0\colon\ \theta_1 = \theta_2 = \cdots = \theta_k = 0.$$

That is, H_0 states that the treatment effect is zero in every study. If we reject H_0 using a combined test procedure, we may safely conclude that H_0 is false. However, H_0 is false if at least one of $\theta_1, \ldots, \theta_k$ is different from zero. Therefore, H_0 could be false when $\theta_1 > 0$ and $\theta_2 = \cdots = \theta_k = 0$. It is doubtful if a researcher would regard such a situation as persuasive evidence of the efficacy of a treatment.

The difficulty in the interpretation of omnibus tests of the significance of combined results stems from the nonparametric nature of the tests. Rejection of the combined null hypothesis allows the investigator to conclude only that the omnibus null hypothesis is false. Errors of interpretation usually involve attempts to attach a parametric interpretation to the rejection of H_0. For example, an investigator might incorrectly conclude that, because H_0 is rejected, the treatment effects are greater than zero (Adcock, 1960). Alternatively, an investigator might incorrectly conclude that the average treatment effect $\bar{\theta}$ is positive. Neither parametric interpretation of the rejection of H_0 is warranted without additional a priori assumptions.

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Omnibus tests of the statistical significance of combined results are poorly suited to the task of drawing general conclusions about the magnitude, direction, and consistency of treatment effects across studies (see Becker, 1987). On the other hand, techniques based on the combination of estimates of effect magnitude do support inferences about direction, magnitude, and consistency of effects. Therefore, statistical analyses based on effect sizes are preferable for most applications for meta-analysis.

Combined Estimation

When all of the studies have similar designs and measure the outcome construct in a similar (but not necessarily identical) manner, the combined estimation approach is probably the preferred method of meta-analysis (Hedges & Olkin, 1985). The first step in combined estimation is the selection of an index of effect magnitude. Many different indexes of effect magnitude have been used in meta-analysis, including the raw mean difference between the treatment and control groups (e.g., Cochran, 1937), the standardized difference between treatment and control groups means (e.g., Smith & Glass, 1977), the correlation coefficient (e.g. Pearson, 1904; Schmidt & Hunter, 1977), the observed minus expected frequency of some outcome like death (e.g., Yusuf, Peto, Lewis, Collins, & Sleight, 1985), the odds ratio between treatment and control groups (Mantel & Haenszel, 1959), or the simple difference between proportions of some outcome in the treatment and control groups (e.g., Devine & Cook, 1983).

Statistical analysis procedures for meta-analysis using any of these indexes of effect magnitude are analogous (Elashoff, 1978; Fleiss, 1973; Gilbert, McPeek, Mosteller, 1977; Hedges, 1983; Hedges & Olkin, 1985; Mantel & Haenszel, 1959; Sheele, 1966). All involve large-sample theory and differ mainly in the details of calculation of standard errors and bias corrections.

Fixed, Random, and Mixed Effects Models

Just as there are three statistical models for the analysis of variance, there are three somewhat analogous conceptualizations of statistical methods in meta-analysis: fixed, random, and mixed effects models. The procedures for data analyses in these models are similar, but the details and interpretation of statistical tests are somewhat different (see Hedges & Olkin, 1985).

In fixed effects conceptualization, the true or population values of the treatment effects are fixed but unknown constants. Usually they are assumed to be a function of study characteristics. By studying the relationship between study characteristics and treatment effects, the data analyst tries to deduce stable relationships that explain essentially all of the variability in study results except for that attributable to within-study sampling variability. The evaluation of particular explanatory models is part of this process.

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However, fixed effect models are not the only way to conceptualize data analysis in meta-analysis.

The random effects conception arises from a model in which the treatment effects are not deterministic functions of known study characteristics. In this model, the true or population values of treatment effects vary randomly from study to study, as if they were sampled from a universe of possible treatment effects (see, DerSimonian & Laird, 1986; Hedges & Olkin, 1985, chap. 9; Hunter & Schmidt, 1990, chap. 3). The random effects conceptualization is consistent with Cronbach's (1980) model for generalization in evaluation studies in which each treatment site (or study) is a sample realization from a universe of related treatments. The primary difference between the interpretation of fixed effects and random effects models is that between-studies variation in treatment effects is conceived to be unsystematic in random-effects models, and consequently explanation of this variance is not possible. Instead, the data analyst usually seeks to quantify this variance by estimating an (treatment by studies) interaction *variance component*: an index of the variability of population treatment effects across studies.

Estimation of variance components to characterize the variability of effect sizes in meta-analysis is necessary for the same reason that the estimation of variance components is necessary in the analysis of variance. Just as sampling error and the variability in treatment effects contribute to mean squares in the analysis of variance, both within-study sampling error and variability in the underlying (population) effect magnitudes contribute to variability in the observed estimates of effect sizes. Because the object of the random-effects analysis is to obtain an estimate of the variability of the underlying effect magnitudes, special procedures are needed (Hedges & Olkin, 1985).

Mixed models involve a combination of the ideas involved in fixed effects and random effects models. In these models, some of the variation between treatment effects is fixed (i.e., explainable), and some is random. Consequently, the data analyst seeks to explain some of the variation between study results and quantify the remainder by estimating a variance component (Raudenbush & Bryk, 1985). Such models have considerable promise as data analytic tools for situations in which it is useful to treat some of the variability between study results as random.

The most important difference in the outcomes produced by the three types of statistical analyses lies in the standard errors that they associate with the overall (combined) estimate of the treatment effect. Fixed effect analyses incorporate only within-study variability into the estimate of the standard error of the combined treatment effect. Random effects analyses typically produce much larger standard errors because they incorporate the

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uncertainty associated with sampling from a superpopulation of studies. Mixed effects analyses incorporate only unexplained between-study variation into the estimate of the standard error of the overall (combined) estimate of the treatment effect and, hence, produce standard errors between those of fixed effects and random effects analyses.

Statistical Methods for Fixed Effects Meta-Analysis

Suppose that $T_1, \ldots, T_k$ are independent estimates of effect magnitude from k studies with sample sizes $n_1, \ldots, n_k$ and unknown population effect magnitude parameters $\theta_1, \ldots, \theta_k$. Assuming that the standard error of T_i is a function of θ_i, denote the standard errors of $T_1, \ldots, T_k$ by $S_1(\theta_1), \ldots, S_k(\theta_k)$ and the estimated standard errors by $S_1(T_1), \ldots, S_k(T_k)$. Assume further that each T_i has a normal asymptotic distribution leading to the large-sample normal approximation

$$T_i \sim N(\theta_i, S^2(T_i)). \tag{6}$$

Estimating the overall average treatment. One of the first statistical questions that arises is how to estimate the overall average treatment effect when it is believed that $\theta_1, \ldots, \theta_k$ are very similar. One way of combining the estimates is obviously to take the simple average of $T_1, \ldots, T_k$. The most precise combination (i.e., the most efficient estimator of θ when $\theta_1 = \cdots = \theta_k = \theta$) is a weighted average that takes the standard errors $S_1(T_1), \ldots, S_k(T_k)$ into account. This weighted average is

$$T_\cdot = \sum_{i=1}^{k} w_i T_i / \sum_{i=1}^{k} w_i, \tag{7}$$

where $w_i = 1/S_i^2(T_i)$. When $N = \sum_{i=1}^{k} n_i \to \infty$ with n_i/N fixed (that is if each study has a large sample size), $T_\cdot$ has an asymptotic distribution leading to the large-sample normal approximation

$$T_\cdot \sim N(\theta, S^2_\cdot(T_\cdot)),$$

where

$$S_\cdot^{-2}(T_\cdot) = \sum_{i=1}^{k} S_i^{-2}(T_\cdot) = \sum_{i=1}^{k} w_i.$$

This result can be used to compute tests of significance and confidence intervals for θ based on $T_\cdot$. A test of the hypothesis that $\theta = 0$ uses the statistic

$$Z = T_\cdot / S_\cdot(T_\cdot),$$

which is compared to the critical values of the standard normal distribution.

Note that, if the $T_i, i = 1, \ldots, k$, are asymptotically efficient (have asymp-

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totic variances equal to the Cramer-Rao lower bound), then $T_.$ is asymptotically efficient.

Another point is that the regularity condition that the n_i/N remain fixed as N approaches infinity is necessary and is also required for all of the other fixed effects analyses discussed in this article. This condition essentially says that each of the experiments becomes large as N approaches infinity. If we let N increase by increasing k and letting the n_i/N remain finite, then a variety of problems arise. For example, under this condition, $T_.$ is not even consistent if the T_i are biased estimators of θ (see, Neyman & Scott, 1948).

Testing homogeneity of treatment effects. Combining estimates of effect magnitude across studies is reasonable if the studies have a common population effect magnitude θ. In this case, the estimates of effect size differ only by unsystematic sampling error. However, if the studies have very different underlying treatment effects, it can be misleading to combine estimates of treatment effects across studies. For example, if half of the studies have a large positive population treatment effect and half of the studies have a negative population treatment effect of equal magnitude, then their average—zero—is not representative of the treatment effects in any of the studies. The obvious question is: How is it possible to determine whether population treatment effects are relatively constant across studies? That is, how do we test for treatment-by-study interactions?

A simple test of homogeneity of treatment effects uses the statistic

$$H_T = \sum_{i=1}^{k} w_i(T_i - T_.)^2, \tag{8}$$

where $w_i = 1/S_i^2(T_i)$ is the weight used in (7) and $T_.$ is the weighted mean given by (7). The H_T statistic is simply the weighted sum of squares of the effect size estimates $T_1, \ldots, T_k$ about the weighted mean of $T_.$. If all studies share a common treatment effect θ, then the statistic H_T has approximately a chi-square distribution with $(k - 1)$ degrees of freedom. Thus, the test for treatment-by-study interaction rejects homogeneity of effect size for large values of H_T.

An analogue to multiple regression analysis for effect magnitudes. If a large or statistically significant value of H_T suggests that the T_i are not estimating the same parameter, the statistician may want to investigate the nature of the differences among the estimates. Often the parameters are assumed to be heterogeneous because of differences among experimental conditions. The object of the analysis in this case may be to examine the effects of these differences on the θ_i. One form this analysis may take is that based on a linear model for the θ_i. Suppose that a vector of $p \leq k$ characteristics $(x_{i1}, \ldots, x_{ip})'$ is known for each experiment and that $\theta_i = \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$, where $\beta_1, \ldots, \beta_p$ are unknown constants. We let $\epsilon_i = T_i - \theta_i$ be the

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sampling error of T_i. Let $\boldsymbol{\theta}$, $\mathbf{T}$, $\boldsymbol{\beta}$, and $\boldsymbol{\epsilon} = (\epsilon_1, \ldots, \epsilon_k)'$ denote the column vectors of parameters, estimates, coefficients, and sampling errors, respectively. That is,

$$\boldsymbol{\theta}' = (\theta_1, \ldots, \theta_k), \quad \mathbf{T}' = (T_1, \ldots, T_k), \quad \text{and}$$
$$\boldsymbol{\beta}' = (\beta_1, \ldots, \beta_p), \quad \text{and} \quad \boldsymbol{\epsilon}' = (\epsilon_1, \ldots, \epsilon_k).$$

Let $\mathbf{X}$ denote a $k \times p$ design matrix of rank p whose ith row is $(x_{i1}, \ldots, x_{ip})$. Thus, the models for $\boldsymbol{\theta}$ and for $\mathbf{T}$ are

$$\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta},$$
$$\mathbf{T} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \tag{9}$$

The problems are to estimate $\boldsymbol{\beta}$, test hypotheses about $\boldsymbol{\beta}$, and assess the specification of the model (9).

A natural estimator of $\boldsymbol{\beta}$ is the generalized least squares estimator

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{V}^{-1}(\mathbf{T})\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}(\mathbf{T})\mathbf{T}, \tag{10}$$

where $\mathbf{V}(\mathbf{T})$ is a $k \times k$ diagonal matrix whose ith diagonal entry is $S_i^2(T_i)$. The asymptotic distribution of $\hat{\boldsymbol{\beta}}$ leads to the large sample normal approximation

$$\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, (\mathbf{X}'\mathbf{V}^{-1}(\mathbf{T})\mathbf{X})^{-1}),$$

which can be used to compute tests and confidence intervals. Note that $\hat{\boldsymbol{\beta}}$ is asymptotically efficient (has asymptotic variance equal to the Cramer-Rao lower bound); thus $\hat{\boldsymbol{\beta}}$ has the same asymptotic distribution as the maximum likelihood estimator of $\boldsymbol{\beta}$.

This result is often used in situations where the model for the data is well understood, but the obvious means of obtaining estimates, such as the method of maximum likelihood, require computation of complicated likelihoods and the use of iterative methods. The present method yields estimators that are simple to compute but have the same large-sample properties as maximum likelihood estimators. In addition, diagnostic procedures routinely used in weighted least squares can be applied in the usual manner.

If the investigator proposes a linear model for $\boldsymbol{\theta}$, it may be desirable to see if the model is reasonably consistent with the estimates $T_1, \ldots, T_k$. If the model does not seem reasonably consistent with the data, then the entire analysis should be suspect. Whenever $k > p$, a natural test of model specification arises in connection with the estimator presented above. The test given below provides a way to check that the data are consistent with the proposed model.

If $\boldsymbol{\beta} = \mathbf{0}$, then the statistic

$$H_M = \mathbf{T}'\mathbf{V}^{-1}(\mathbf{T})\mathbf{X}(\mathbf{X}'\mathbf{V}^{-1}(\mathbf{T})\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}(\mathbf{T})\mathbf{T} \tag{11}$$

is distributed approximately as a chi-square with p degrees of freedom. The

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statistic H_M can be used for a simultaneous test that $\beta_1 = \cdots = \beta_p = 0$. More significantly, H_M is used in the calculation of a test for goodness of fit, or specification of the linear model. If $k > p$ and if the model $\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta}$ is correctly specified, then the statistic

$$H_E = \mathbf{T}'\mathbf{V}^{-1}(\mathbf{T})\mathbf{T} - H_M \tag{12}$$

has an approximate distribution given by

$$H_E \sim \chi^2_{k-p}.$$

The statistic H_M is the (weighted) sum of the squares due to the regression model, and the statistic H_E is the (weighted) sum of squares about the regression plane (the error sum of squares). Thus the test for model misspecification is a test for larger than expected residual variation. If H_E is large or significant, the investigator might use any of the standard tools of regression analysis (e.g., the examination of residuals, the search for influential observation, etc.) to look for problems with the model.

Statistical Methods for Random Effects Meta-Analysis

Again suppose that $T_1, \ldots, T_k$ are independent estimates of treatment effects from k experiments with (unknown) population treatment effects $\theta_1, \ldots, \theta_k$. Again denote the standard error of T_i given by θ_i by $S_i(\theta_i)$, and denote the estimated standard error of T_i by $S_i(T_i)$. Assume also that the T_i are approximately normally distributed about the corresponding θ_i. Now, however, introduce the random effects model that $\theta_1, \ldots, \theta_k$ are sampled from a hyperpopulation of treatment effects. Often the θ_is are assumed to be normally distributed, but this is not essential. The object of the analysis is to estimate the mean $\bar{\theta}$ and variance σ^2_θ (the hyperparameters) of the distribution of population treatment effects and to test the hypothesis that $\bar{\theta} = 0$.

Testing the variance component. A test of the hypothesis that the between-studies variance component σ^2_θ is equal to zero uses the homogeneity statistic H_T given in (8). This test is identical to the test of homogeneity of effect sizes in the fixed effects model, since if $\sigma^2_\theta = 0$ the treatment effect magnitudes are identically equal to zero.

Estimating the variance component. A distribution free approach to estimating σ^2_θ is analogous to the procedure used to estimate the variance component in the one-factor random effects analysis of variance. One simple estimate σ^2_θ is given by

$$\hat{\sigma}^2_\theta = S^2_T - \sum_{i=1}^{k} S^2_i(T_i)/k, \tag{13}$$

where S^2_T is the usual sample variance of $T_1, \ldots, T_k$ and values smaller than zero are usually truncated at zero (see Hedges & Olkin, 1985).

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A slight variation on the approach is to use the weighted variance H_T given in (8) in constructing the estimate of σ_θ^2 (e.g., see DerSimonian & Laird, 1986). This estimate is given by

$$\hat{\sigma}_\theta^2 = \frac{H_T - (k-1)}{c}, \tag{14}$$

where

$$c = \sum_{i=1}^{k} w_i - \left[\sum_{i=1}^{k} w_i^2\right] \Big/ \left[\sum_{i=1}^{k} w_i\right]$$

and values smaller than zero are usually truncated at zero. Because it is based on the statistic used for testing the significance of σ_θ^2, the estimate given in (14) of σ_θ^2 has the advantage over that given in (13) of always yielding estimates that are qualitatively consistent with the significance test. That is, estimates given by (13) are larger than zero only if H_T exceeds its expected value $(k-1)$ and are always positive when the test declares σ_θ^2 significantly greater than zero. It is possible (but quite unusual) for the estimate of σ_θ^2 based on (13) to be zero even when the test declares σ_θ^2 to be significantly greater than zero.

Estimating the mean effect. The usual estimate of the weighted mean is

$$T_\cdot^* = \sum_{i=1}^{k} w_i^* T_i \Big/ \sum_{i=1}^{k} w_i^*, \tag{15}$$

where $w_i^* = 1/[\hat{\sigma}_\theta^2 + S_i^2(T_i)]$. If the random effects are normally distributed, the weighted mean $T_\cdot^*$ is approximately distributed

$$T_\cdot^* \sim N(\bar{\theta}, \sigma_*^2),$$

where

$$\sigma_*^{-2} = \sum_{i=1}^{k} w_i^*. \tag{16}$$

Note that the weights w_i^* used in (15) are not the same as the weights w_i used in (7) unless $\hat{\sigma}_\theta^2$ is exactly zero. Usually σ_θ is larger than zero, and consequently $T_\cdot^*$ may differ from $T_\cdot$. Moreover, the standard error σ_* of $T_\cdot^*$ is usually larger (often much larger) than the standard error $\sigma_\cdot$ of $T_\cdot$. As a result, overall treatment effects that are significantly different from zero in a fixed effect analysis may not be significant in a random effects analysis. The difference, of course, results from differences in the conceptualization of the model and in what counts as sampling error.

Parametric procedures. More complex procedures for estimating σ_θ^2 under various distributional assumptions on the θ_i and for estimating testing hypotheses about $\bar{\theta}$ are given in Rubin (1980, 1981), Raudenbush and Bryk

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(1985), and Hedges and Olkin (1985). Although they involve stronger assumptions, these procedures have the advantage of taking into account the uncertainty of the estimate of σ_θ^2 in analyses of $\bar{\theta}$.

Statistical Methods for Mixed Effects Meta-Analyses

Assume as in earlier models that $T_1, \ldots, T_k$ are independent estimates of treatment effects from k experiments with (unknown) population treatment effects $\theta_1, \ldots, \theta_k$. Denote the standard error of T_i given θ_i by $S_i(\theta_i)$ and its estimate based on T_i as $S_i(T_i)$. Assume that the T_i are normally distributed about the corresponding θ_i. The usual specification of the mixed model is that the θ_i are determined partially by a set of fixed explanatory variables. Suppose that a vector of $p \leq k$ characteristics $(x_{i1}, \ldots, x_{ip})'$ is known for each experiment and that

$$\theta_i = \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \eta_i,$$

where $\beta_1, \ldots, \beta_p$ are unknown constants and η_i is a residual taken to vary randomly between studies. We could also denote a sampling error $\epsilon_i = T_i - \theta_i$ for each study. Let $\boldsymbol{\theta}$, $\mathbf{T}$, $\boldsymbol{\beta}$, η, and $\boldsymbol{\epsilon}$ denote column vectors of parameters, estimates, coefficients, sampling errors, and random effects, respectively. That is,

$$\boldsymbol{\theta}' = (\theta_1, \ldots, \theta_k), \quad \mathbf{T}' = (T_1, \ldots, T_k), \quad \boldsymbol{\beta}' = (\beta_1, \ldots, \beta_p),$$

$$\boldsymbol{\eta}' = (\eta_1, \ldots, \eta_k), \quad \text{and} \quad \boldsymbol{\epsilon}' = (\epsilon_1, \ldots, \epsilon_k).$$

Let $\mathbf{X}$ denote a $k \times p$ design matrix of rank p in which the ith row is $(x_{i1}, \ldots, x_{ip})$. Thus, the model for $\boldsymbol{\theta}$ is

$$\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\eta},$$

and the model for $\mathbf{T}$ is

$$\mathbf{T} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\eta} + \boldsymbol{\epsilon}. \tag{17}$$

The general problem is to estimate and test hypotheses about $\boldsymbol{\beta}$ and to estimate the variance of the η_i.

Testing the variance component. A test of the hypothesis that the variance σ_η^2 of the effects is equal to zero uses the model specification statistic H_M given in (11). This test is identical to the test that is used to determine if the fixed effects model is correctly specified. The reason is that if $\sigma_\eta^2 = 0$ the effect sizes are just a linear combination of the predictor variables and Model 17 reduces to (9).

Estimation. As in the case of random effects models, there are two distinct strategies for obtaining estimates and statistical tests. The distribution free approach uses a relatively simple distribution free estimate of the variances of the η_is and then estimates $\boldsymbol{\beta}$ via generalized least squares (see

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Hanushek, 1974). A distributional assumption on the η_is is invoked only when hypothesis tests are desired. One estimator of the variance of η is

$$\hat{\sigma}_\eta^2 = (\mathbf{T}'\mathbf{T} - \mathbf{T}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{T})/(k-p) - \sum_{i=1}^{k} S_i^2(T_i)/k,$$

which leads to the estimate of $\boldsymbol{\beta}$ given by

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{T},$$

where $\mathbf{V}$ is a $k \times k$ diagonal matrix in which the ith diagonal entry is

$$S_i^2(T_i) + \hat{\sigma}_\eta^2.$$

The other approach to mixed model analyses is to specify a distributional form for the random effects η_i and use this distributional specification explicitly in both estimation and testing (e.g., via maximum likelihood). This second approach is more involved and is usually carried out by considering the meta-analysis in the framework of a general multilevel statistical model.

Statistical methods for mixed effects meta-analyses have received less complete treatment in the literature than have fixed and random effects models. There is a great deal of work in progress on meta-analysis with mixed effects models (see Raudenbush, in press). This work shows a great deal of potential for resolving differences between fixed and random effects approaches. Useful treatments of mixed effects meta-analyses in the context of educational research are given in Raudenbush and Bryk (1985) and Becker (1988).

Areas of Current Research

Applications of the methods described in this article frequently encounter problems that arise because real collections of research studies exhibit complications not present in the simplest conceptualizations of meta-analysis. Methodological research in meta-analysis has been largely driven by such practical problems (see Hedges, 1990).

Perhaps the most pervasive problem in large scale meta-analyses is that of missing data. Reporting of experimental conditions and statistical analyses is rarely complete enough to permit extraction of usable data from all relevant studies. In some cases, such as the meta-analysis of test validity studies, a very large proportion of the studies typically contains missing data (e.g., data on restriction of range or unreliability needed to render reliability coefficients comparable across studies). The development of more effective ways to carry out meta-analyses involving missing data is an area of active research interest (e.g., see Hedges, 1988; Hunter & Schmidt, 1990).

Publication bias or the file drawer problem is another major problem in meta-analysis. It arises when data from some studies are not available for

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synthesis and the probability that the data are available is related to the size of the effect. Typically publication or reporting bias arises because studies that find statistically significant results are often more likely to be submitted for publication, to be accepted for publication, and to have their results reported in sufficient detail to permit extraction of an effect size estimate (see, e.g., Begg & Berlin, 1988). One promising approach to the problem involves modeling the selection process as part of the estimation procedure (see Hedges, 1992; Iyengar & Greenhouse, 1988).

A third problem that has stimulated considerable research is that of combining estimates of effect size that are not independent. Often such estimates arise when study designs permit (or require) several estimates of effect size to be computed from the same study. Examples include experimental studies that measure several outcomes of the same type or have multiple treatment or control groups and correlational studies that measure several relationships using the same individuals (see Becker, 1992; Gleser & Olkin, in press; Hedges & Olkin, 1985, chap. 10; Raudenbush, Becker, & Kalaian, 1988).

A final problem arises in the context of random effects analyses. Distribution free approaches to estimation are rather crude, and assumptions about the distribution of random effects are usually invoked during testing in any event. Yet we usually have little insight about the form that the distribution of random effects might take. Normality is one possibility, but it is one that is invoked more often because of convenience than because of reasoned argument. Analyses involving estimation of the form of the distribution of the random effects are one promising line of research that can improve on the distribution free or the fully parametric approaches (see, e.g., Laird, 1978; Laird, 1982; Laird & Louis, 1987).

References

Adcock, C. J. (1960). A note on combining probabilities. *Psychometrika*, *25*, 303–305.

Becker, B. J. (1987). Applying tests of combined significance in meta-analysis. *Psychological Bulletin*, *102*, 164–171.

Becker, B. J. (1988). Synthesizing standardized mean-change measures. *British Journal of Mathematical and Statistical Psychology*, *41*, 257–278.

Becker, B. J. (1992). Models of science achievement: Factors affecting male and female performance in school science. In T. Cook, H. Cooper, D. Cordray, L. Hedges, R. Light, T. Louis, & F. Mosteller, *Towards explanatory meta-analysis* (pp. 209–281). New York: Russell Sage Foundation.

Begg, C., & Berlin, J. A. (1988). Publication bias: A problem in interpreting medical research. *Journal of the Royal Statistical Society*, *Series A*, *151*, 1–27.

Birge, R. T. (1932). The calculation of errors by the method of least squares. *Physical Review*, *16*, 1–32.

Downloaded from http://jebs.aera.net at MICHIGAN STATE UNIV LIBRARIES on February 12, 2015

Cochran, W. C. (1937). Problems arising in the analysis of a series of similar experiments. *Journal of the Royal Statistical Society Supplement*, *4*, 102–118.

Cronbach, L. J. (1980). *Toward reform of program evaluation*. San Francisco: Jossey-Bass.

DerSimonian, R., & Laird, N. (1986). Meta-analysis in clinical trials. *Controlled Clinical Trials*, *7*, 177–188.

Divine, E. C., & Cook, T. D. (1983). A meta-analytic review of psychoeducational interventions on length of postsurgical hospital stay. *Nursing Research*, *32*, 267–274.

Elashoff, J. D. (1978). Combining the results of clinical trials. *Gastroenterology*, *28*, 1170–1172.

Fisher, R. A. (1932). *Statistical methods for research workers* (4th ed.). London: Oliver & Boyd.

Fleiss, J. L. (1973). *Statistical methods for rates and proportions*. New York: Wiley.

George, E. O. (1977). *Combining independent one-sided and two-sided statistical tests—Some theory and applications*. Unpublished doctoral dissertation, University of Rochester.

George, E. O., & Mudholkar, G. S. (1977). The logit method for combining independent tests. *Institute of Mathematical Statistics Bulletin*, *6*, 212.

Gilbert, J. P., McPeek, B., & Mosteller, F. (1977). Progress in surgery and anesthesia: Benefits and risks of innovation therapy. In J. Bunker, B. Barnes, & F. Mosteller (Eds.), *Costs, risks, and benefits of surgery* (pp. 170–175). New York: Oxford University Press.

Glass, G. V. (1976). Primary, secondary, and meta-analysis of research. *Educational Researcher*, *5*(10), 3–8.

Gleser, L., & Olkin, I. (in press). Multiple stochastically dependent outcomes. In H. Cooper & L. V. Hedges (Eds.), *The handbook of research synthesis*. New York: Russell Sage Foundation.

Hanushek, E. A. (1974). Efficient estimators for regression coefficients. *American Statistician*, *28*, 66–67.

Hedges, L. V. (1983). Combining independent estimators in research synthesis. *The British Journal of Mathematical and Statistical Psychology*, *36*, 123–131.

Hedges, L. V. (1988). Improving statistical procedures for validity generalization. In H. Braun & H. Wainer (Eds.), *Test validity for the 1990s and beyond* (pp. 191–212). Hillsdale, NJ: Erlbaum.

Hedges, L. V. (1990). The future of statistical methods for meta-analysis. In M. Straf & K. Wachter (Eds.), *The future of meta-analysis* (pp. 11–26). New York: Russell Sage Foundation.

Hedges, L. V. (1992). Modeling publication selection effects in meta-analysis. *Statistical Science*, *7*, 246–255.

Hedges, L. V., & Olkin, I. (1985). *Statistical methods for meta-analysis*. New York: Academic.

Hunter, J. E., & Schmidt, F. L. (1990). *Methods of meta-analysis: Correcting error and bias in research findings*. Newbury Park, CA: Sage.

Iyengar, S., & Greenhouse, J. B. (1988). Selection models and the file drawer problem. *Statistical Science*, *3*, 109–135.

Downloaded from http://jebs.aera.net at MICHIGAN STATE UNIV LIBRARIES on February 12, 2015

Laird, N. M. (1978). Nonparametric maximum likelihood estimate of a mixing distribution. *Journal of the American Statistical Association*, *73*, 805–811.

Laird, N. M. (1982). Empirical Bayes estimate using the nonparametric maximum likelihood estimate of the prior. *Journal of Statistical Computing and Simulation*, *1*, 211–220.

Laird, N. M., & Louis, T. M. (1987). Empirical Bayes estimate based on bootstrap samples. *Journal of the American Statistical Association*, *82*, 739–750.

Mantel, N., & Haenszel, W. (1959). Statistical aspects of the analysis of data from retrospective studies. *Journal of the National Cancer Institute*, *22*, 719–748.

Mudholkar, G. S., & George, E. O. (1979). The logit method for combining probabilities. In J. Rustagi (Ed.), *Symposium on optimizing methods in statistics* (pp. 345–366). New York: Academic.

Neyman, J., & Scott, E. L. (1948). Consistent estimates based on partially consistent observations. *Econometrica*, *16*, 1–32.

Pearson, K. (1904). Report on certain enteric fever inoculation statistics. *British Medical Journal*, *2*, 1243–1246.

Pearson, K. (1933). On a method of determining whether a sample of given size *n* supposed to have been drawn from a parent population having a known probability integral has probably been drawn at random. *Biometrika*, *25*, 379–410.

Raudenbush, S. W. (in press). Random effects models. In H. M. Cooper & L. V. Hedges (Eds.), *The handbook of research synthesis*. New York: Russell Sage Foundation.

Raudenbush, S. W., Becker, B. J., & Kalaian, H. (1988). Modeling multivariate effect sizes. *Psychological Bulletin*, *103*, 111–120.

Raudenbush, S. W., & Bryk, A. S. (1985). Empirical Bayes meta-analysis. *Journal of Educational Statistics*, *10*, 75–98.

Rubin, D. B. (1980). Using empirical Bayes techniques in the law school validity studies. *Journal of the American Statistical Association*, *75*, 801–816.

Rubin, D. B. (1981). Estimation in parallel randomized experiments. *Journal of Educational Statistics*, *6*, 337–401.

Schmidt, F., & Hunter, J. (1977). Development of a general solution to the problem of validity generalization. *Journal of Applied Psychology*, *65*, 643–661.

Sheele, P. R. (1966). Combination of log-relative risks in retrospective studies of disease. *American Journal of Public Health*, *56*, 1745–1750.

Smith, M. L., & Glass, G. V. (1977). Meta-analysis of psychotherapy outcome studies. *American Psychologist*, *32*, 752–760.

Stigler, S. M. (1986). *The history of statistics: The measurement of uncertainty before 1900*. Cambridge, MA: Harvard University Press.

Stouffer, S. A., Suchman, E. A., DeVinney, L. C., Star, S. A., & Williams, R. M., Jr. (1949). *The American soldier: Vol. 1. Adjustment during Army life*. Princeton: Princeton University Press.

Tippett, L. H. C. (1931). *The method of statistics*. London: Williams & Norgate.

Wallis, W. A. (1942). Compounding probabilities from independent significance tests. *Econometrica*, *10*, 229–248.

Yusuf, S., Peto, R., Lewis, J., Collins, R., & Sleight, P. (1985). Beta blockade during and after myocardial infarction: An overview of randomized trials. *Progress in Cardiovascular Diseases*, *27*, 335–371.

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